

NOVABC BANK

Customer Churn Analytics

Identifying Why Customers Leave — and Who's Next

Analytics Report

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Pipeline: Databricks Lakehouse (Medallion Architecture) · Power BI

80,000 customer records analysed

EXECUTIVE SUMMARY

Executive Summary

NovaBC Bank loses 18% of its customer base annually — 14,400 of 80,000 customers churned in the analysed period. This report moves beyond reporting how much churn occurred to identify why, using a behavioural analytics pipeline built on Databricks and visualised in Power BI.

The central finding: disengagement, not financial risk, is the dominant churn driver. The bank's existing risk framework only ever classifies customers as Low or Medium risk and shows no meaningful link to actual churn outcomes. A custom, independently-engineered risk label built from behavioural signals — activity status and engagement score — proved four times more predictive, and revealed that 34,400 currently active customers show the identical warning pattern that already caused 12,266 customers to leave.

80,000 CUSTOMERS ANALYSED	18% OVERALL CHURN RATE	34,400 AT-RISK, NOT YET CHURNED	4x HIGH VS LOW RISK CHURN
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Key Takeaways

- Disengagement is the single largest churn driver, and it is still spreading — 43% of the entire customer base shows the same behavioural pattern as customers who already left.
- Inactive members churn at 3x the rate of active members (21.1% vs 6.5%).
- Low-balance customers churn at nearly 20x the rate of Premium-balance customers (38% vs 2%).
- Life-stage segments (students, retirees, general labour) show both the lowest engagement and highest churn — occupation is a proxy for engagement, not a direct cause.
- Tenure offers only modest protection against churn compared to engagement and risk signals.

BACKGROUND

Business Problem & Methodology

The Problem

The bank's existing risk framework (risk_score / risk_segment) classifies customers on financial grounds only, and produces no High Risk category at all — every customer is labelled Low or Medium. This leaves the bank with no systematic way to identify which customers are about to leave, why, or which currently active customers show the same warning signs.

Approach

Data was processed through a Bronze → Silver → Gold Medallion pipeline on Databricks, then visualised in a two-page Power BI dashboard.

Layer	What Happened
Bronze	80,000 raw records ingested to Delta Lake with full data lineage tracking. Zero transformations.
Silver	Data cleaned, PII removed, modelled into a star schema (1 fact + 3 dimension tables), and enriched with 6 engineered features derived from actual data distributions.
Gold	8 KPI tables built with Spark SQL (CTEs, UNION ALL, window functions) aggregating churn across every business dimension.
Power BI	Gold tables imported (Import mode) into a 2-page executive dashboard for reporting and root-cause analysis.

Engineered Features

Six features were engineered in the Silver layer. Two are original, independently-validated churn signals not present in the source data:

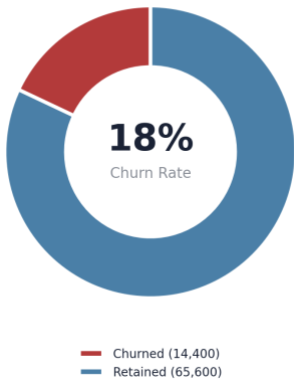
Feature	Logic	Purpose
balance_tier	Quartile cutoffs (25th/50th/75th pctl)	Ensures even distribution across financial segments
engagement_tier	Low/Med/High from score distribution	Turns raw 0-100 score into an actionable segment
tenure_group	New/Regular/Loyal (data-driven cutoffs)	Avoids meaningless categories given max tenure = 4 yrs
churn_risk_label *	Inactive + low engagement -> High Risk	Independent behavioural risk score, validated at 4x predictive power
churn_reason_inferred *	Rule-based, applied to churned AND active customers	Identifies why customers left and flags at-risk customers before they leave

* Custom-engineered features, independently validated against actual churn outcomes.

FINDINGS

Key Findings

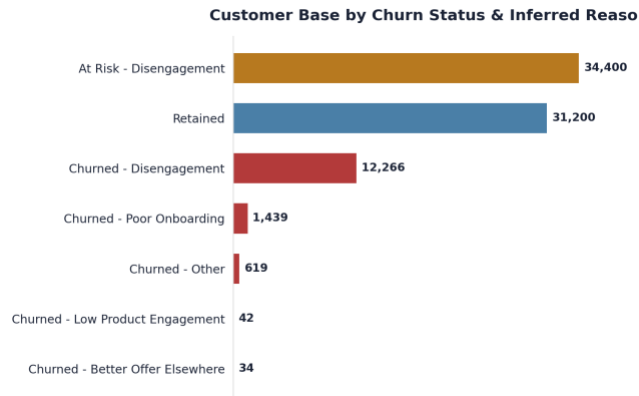
FINDING 1 Overall Churn Snapshot



Metric	Value
Total Customers	80,000
Churned Customers	14,400
Churn Rate	18%
Average Balance	61,183,195 VND
Average Engagement Score	31.73 / 100

FINDING 2 Disengagement Is the Leading Driver — And It's Still Growing

This is the most actionable finding in the analysis. 12,266 customers have already left while inactive and disengaged. A further 34,400 currently active customers show the exact same pattern — meaning 43% of the entire customer base carries the same risk profile that already produced real churn.



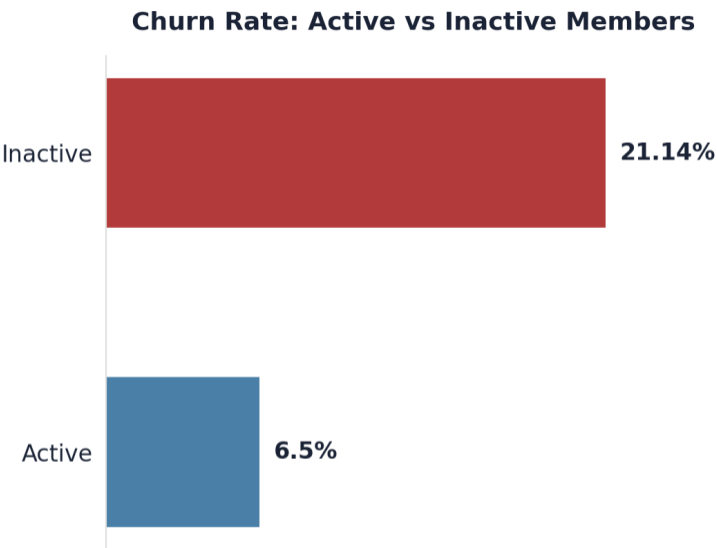
Customer base segmented by churn status and inferred reason (source: gold_churn_by_reason)

WHY THIS MATTERS

These 34,400 customers have not churned yet. Unlike the 12,266 who already left, this is a live retention window — the single largest opportunity identified in this analysis.

FINDING 3 Inactivity Triples Churn Risk

Member activity status is the single strongest behavioural predictor of churn. Inactive members churn at three times the rate of active members, with average engagement scores more than three times lower.

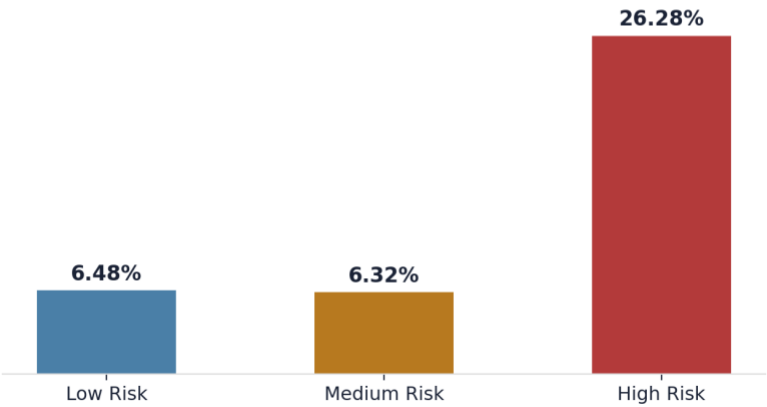


Member Status	Total Customers	Churned	Churn Rate	Avg Engagement
Inactive	62,824	13,283	21.14%	20.37
Active	17,176	1,117	6.50%	73.25

FINDING 4 Custom Risk Label Outperforms the Bank's Existing Risk Score

The dataset's built-in risk_score and risk_segment measure general financial risk and classify nearly all customers as Low or Medium — with no meaningful separation by actual churn outcome. A custom churn_risk_label was engineered independently, using only behavioural signals (activity status + engagement tier), deliberately excluding the existing risk_score to test whether behaviour alone carries predictive signal.

Churn Rate by Custom Risk Label (Validation)



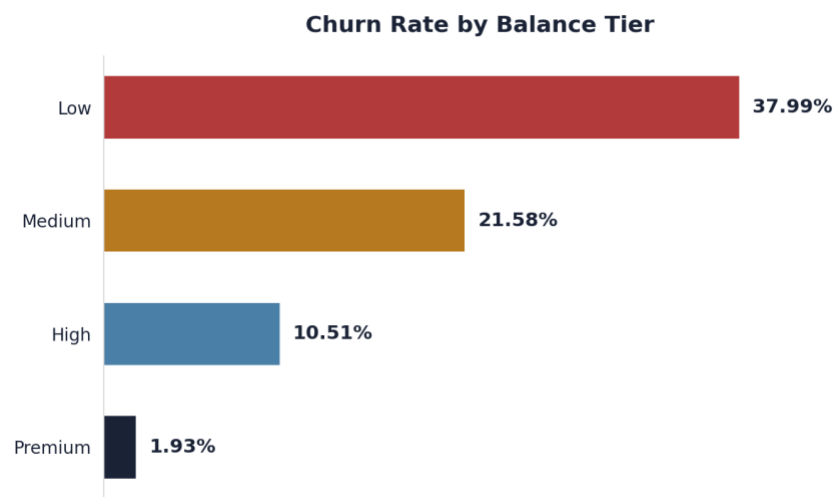
Risk Label	Total Customers	Churned	Churn Rate	Avg Engagement	Avg Risk Score
High Risk	46,666	12,266	26.28%	16.29	0.32
Medium Risk	16,191	1,023	6.32%	32.19	0.22
Low Risk	17,143	1,111	6.48%	73.30	0.18

VALIDATION RESULT

High Risk customers churn at 4x the rate of Low Risk customers — confirming the behavioural label carries real predictive signal the bank's existing financial risk score does not capture.

FINDING 5 Financial Profile Still Matters — Balance Is the Strongest Financial Predictor

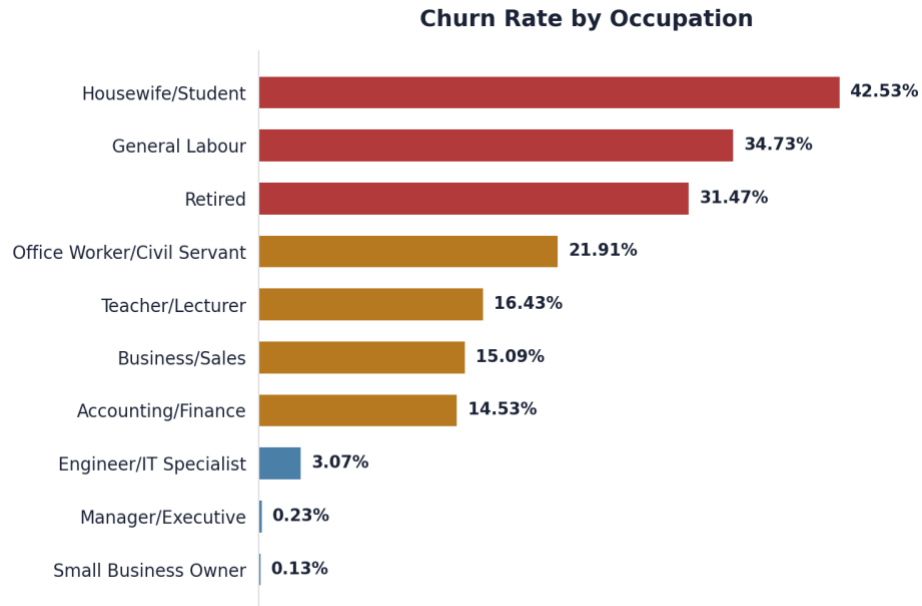
Balance tiers were built using quartile cutoffs from the actual balance distribution (25th/50th/75th percentile) rather than arbitrary round numbers, ensuring each tier holds a comparable share of customers. The result is the strongest single financial predictor in the dataset.



Balance Tier	Total Customers	Churned	Churn Rate
Low	19,994	7,596	37.99%
Medium	20,001	4,316	21.58%
High	19,998	2,101	10.51%
Premium	20,007	387	1.93%

FINDING 6 Life-Stage Segments Show the Widest Churn Gap

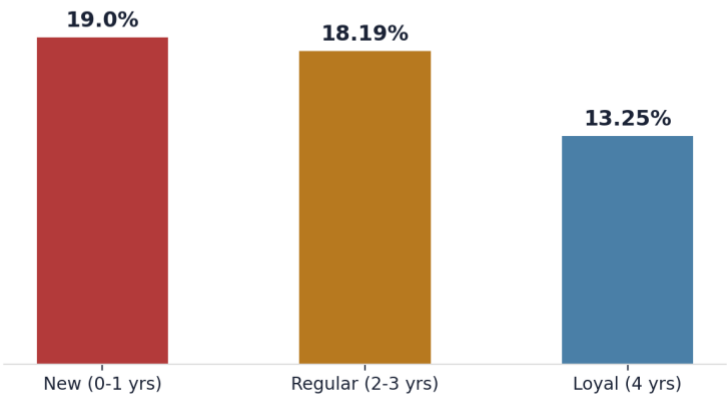
Occupation reveals the largest spread of any segment in the dataset — from 42.5% churn among students/housewives down to 0.13% among small business owners. The pattern tracks average engagement score almost exactly, suggesting occupation is a proxy for engagement rather than a direct cause of churn.



FINDING 7 Tenure Offers Only Modest Protection

Customer relationship length has a real but comparatively weak effect on churn — a 5.75-point gap between New and Loyal customers, far smaller than the 3-4x swings seen with engagement and risk label. This suggests that simply waiting for customers to age into loyalty is a much weaker retention lever than actively managing engagement.

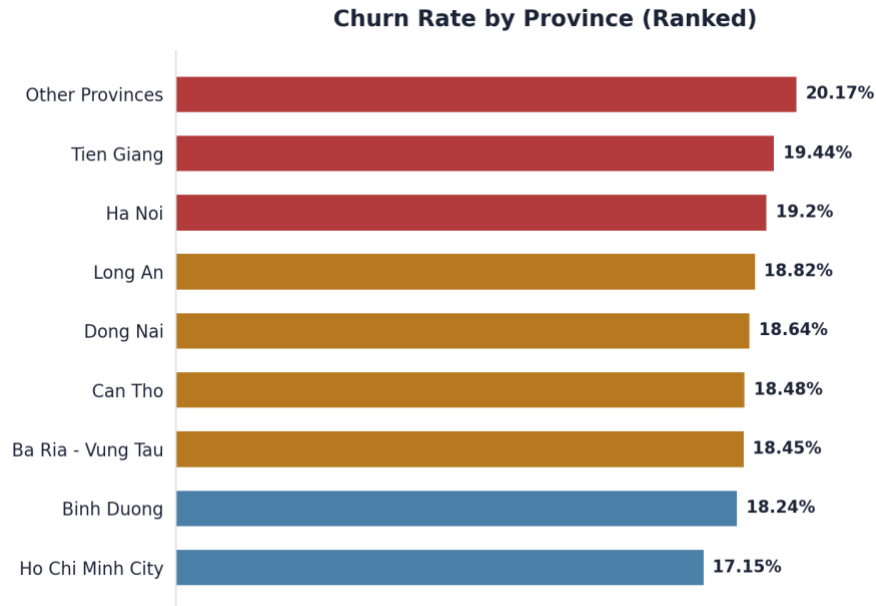
Churn Rate by Tenure Group



Tenure Group	Total Customers	Churned	Churn Rate
New (0-1 yrs)	35,677	6,780	19.00%
Regular (2-3 yrs)	35,386	6,436	18.19%
Loyal (4 yrs)	8,937	1,184	13.25%

FINDING 8 Geographic Distribution Is Broad, Not Concentrated

Provinces were ranked by churn rate using a RANK() window function in Spark SQL. Churn is broadly distributed across the customer base rather than concentrated in a single region — meaning geographic targeting alone would miss the majority of at-risk customers.



Top: 'Other Provinces' (grouped smaller regions), 20.17% churn. Bottom: Ho Chi Minh City — largest customer base (40,148 customers) — 17.15% churn, rank 9 of 9.

DASHBOARD

Power BI Dashboard

The findings above are consumed through a two-page interactive Power BI dashboard. Page 1 answers what is happening with churn; Page 2 answers why, and identifies which currently active customers are showing early warning signs.

Page 1 — Executive Overview

KPI cards (Total Customers, Churned, Churn Rate, Retention Rate, Avg Balance) · Churn by Age Group · Province Performance · Activity Impact · Churn Risk Scatter · Churn by Balance Tier · Slicers: Gender, Age Group

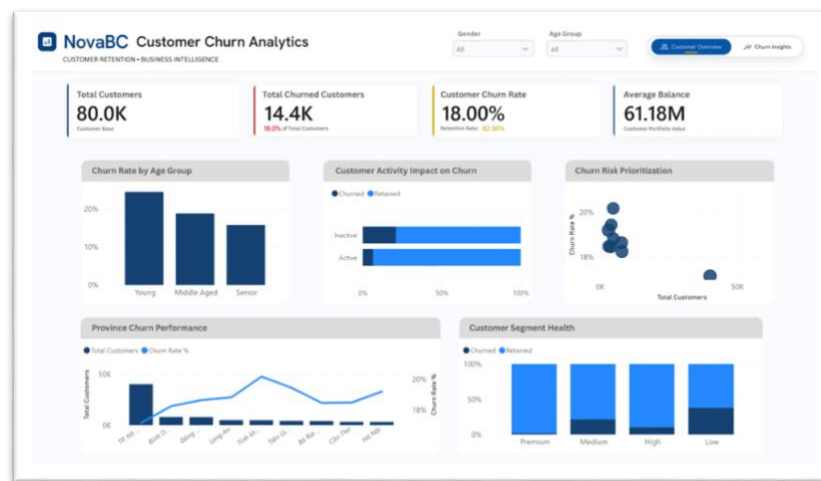


Figure — Power BI Dashboard, Page 1: Executive Overview

Page 2 — Churn Risk & Root Cause Analysis

KPI cards (High Risk, At Risk, Avg Engagement, High Risk Churn Rate) · Churn by Inferred Reason · Churn by Risk Label · Occupation Risk Treemap · Engagement Score Distribution · Risk vs Engagement Heatmap Matrix · Slicers: Churn Risk Label, Occupation

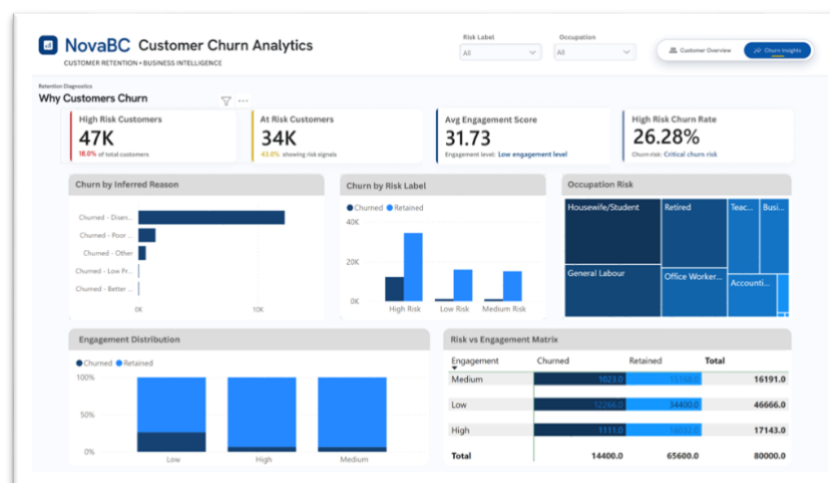


Figure — Power BI Dashboard, Page 2: Churn Risk & Root Cause Analysis

RECOMMENDATIONS

Recommendations

1. Prioritise the 34,400 at-risk customers immediately.

This is the highest-leverage group identified in the analysis — behaviourally identical to customers who already churned, but still retainable. An intervention here has a direct, measurable path to reducing next-period churn.

2. Build engagement-based early-warning triggers, not just financial ones.

Activity status and engagement score predict churn 3-4x more effectively than the bank's existing financial risk score. An inactivity-based alert system would catch risk the current framework misses entirely.

3. Target low-balance and life-stage-transitional segments with tailored engagement programmes.

Students, retirees, and general labour customers show both the lowest engagement and highest churn. Generic offers are unlikely to move this segment — engagement-first outreach is more consistent with the underlying driver.

4. Reframe the existing risk_score as a financial-health metric only.

It should not be relied upon as a churn indicator on its own — pairing it with behavioural data (as this analysis does) closes a real blind spot in the current framework.

APPENDIX

Limitations & Notes

- churn_reason_inferred is a rule-based approximation, not a customer-reported reason. The source data does not capture stated reasons for leaving, so this should be treated as a hypothesis for further validation (e.g. customer surveys) rather than ground truth.
- The dataset represents a single snapshot; churn drivers may shift seasonally or in response to external factors (rate changes, competitor offers) not captured here.
- Currency values are reported in Vietnamese Dong (VND) and were not converted, as no reliable exchange-rate timestamp was available for the data collection period.

Technology Stack

Layer	Tool
Platform	Databricks Lakehouse, Unity Catalog
Processing	PySpark, Spark SQL
Storage	Delta Lake
Modeling	Star Schema (SQL)
Visualisation	Power BI (Import mode)

Gold Layer KPI Tables

Table	SQL Technique
gold_churn_summary	Aggregation
gold_churn_by_segment	CTE + UNION ALL
gold_churn_by_reason	Aggregation
gold_churn_risk_analysis	Aggregation
gold_churn_by_occupation	Aggregation
gold_churn_active_vs_inactive	Aggregation
gold_churn_by_tenure	Aggregation
gold_churn_top_provinces	Window Function RANK()